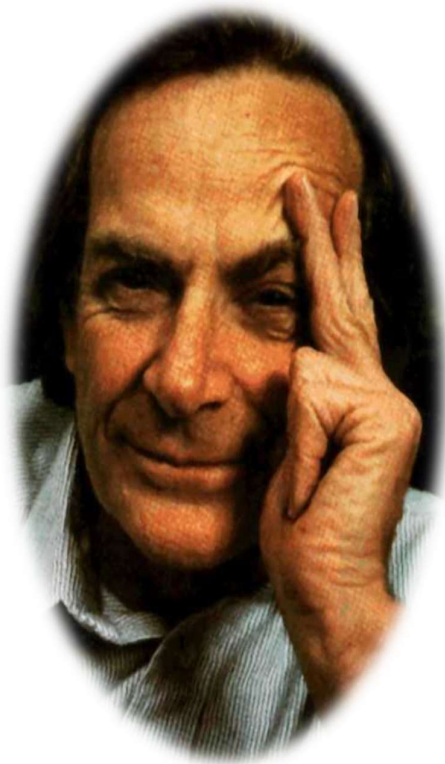


The Value of Science

- *Richard P. Feynman*



Richard P. Feynman was born in New York City on the 11th of May 1918. He studied at the Massachusetts Institute of Technology where he obtained his B.Sc. in 1939 and at Princeton University where he obtained his Ph.D. in 1942. He was Research Assistant at Princeton (1940-1941), Professor of Theoretical Physics at Cornell University (1945-1950), Visiting Professor, and thereafter appointed

Professor of Theoretical Physics at the California Institute of Technology (1950-1959). At present he is Richard Chace Tolman Professor of Theoretical Physics at the California Institute of Technology.

Professor Feynman is a member of the American Physical Society, the American Association for the Advancement of Science; the National Academy of Science; in 1965 he was elected a foreign member of the Royal Society, London (Great Britain).

He holds the following awards: Albert Einstein Award (1954, Princeton); Einstein Award (Albert Einstein Award College of Medicine); Lawrence Award (1962).

Richard Feynman is married to Gweneth Howarth, they have a son, Carl Richard (born 22nd April 1961), and a daughter Michelle Catherine (born 13th August 1968).

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Richard Feynman's Four Greatest Achievements In Physics



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“When I was younger, I thought science would make good things for everybody. It was obviously useful; it was good. During the war I worked on the atomic bomb. This result of science was obviously a very serious matter: it represented the destruction of people. After the war I was very worried about the bomb. I didn't know what the future was going to look like, and I certainly wasn't anywhere near sure that we would last until now. Therefore, one question was—is there some evil involved in science? Put another way—what is the value of the science I had dedicated myself to—the thing I loved—when I saw what terrible things it could do? It was a question I had to answer. —The Value of Science is a kind of report, if you will, on many of the thoughts that came to me when I tried to answer that question.”
—Richard Feynman

The Value of Science — through Feynman's lens.

Following his involvement in the Manhattan Project Richard Feynman suffered an existential crisis in which he questioned the fate of humanity and deeply considered the evil that could be created by science. Through these thoughts, he arrived at various conclusions of what the value of science is. These conclusions were presented to a host of great scientific minds at Caltech University in the summer of 1955.

I. Scientific knowledge is a power that allows us to do and make things

Scientific knowledge and the vast expanse that it is, allows us to do a create a multitude of things. The problem is that this power is not accompanied by any instructions. The lack of instructions presents a dilemma, one which Feynman deals with through reference to a Buddhist quote which reads:

“To every man is given the key to the gates of heaven; the same key opens the gates of hell.”

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A lack of instruction places the key holder in a precarious position, as that keyholder could very easily open the wrong gate. The dilemma occurs because, despite the danger of having no instructions, it is impossible to enter heaven without the keys, and such, instructions without a key are worthless.

Science possesses the ability to produce horror, yet it is of value simply because it can produce something.

II. Entertainment and Inspirational Value

Some get this intellectual enjoyment from reading, writing or learning about science, while others get it from working with science. Either way, it is important that science is given due recognition for its educational and inspirational value. Science has allowed us to imagine things infinitely more marvellous than poets and dreamers of the past. Feynman portrays this idea beautifully and provides perspective as he asks, ‘how much more remarkable it is for us all to be stuck — half of us upside down — by a mysterious attraction to a

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spinning ball that has been swinging in space for billions of years than to be carried on the back of an elephant supported on a tortoise swimming in a bottomless sea?" He goes on to give the listeners a peek into his mind and his thoughts and the questions that he has about our world in a paragraph that can be read over and over and never lose its elegance or its ability to create bewilderment.

"The imagination of nature is far, far greater than the imagination of man."

This cycle of intellectual and enjoyment is formed by a never-ending cycle of curiosity, one which is possible because with more knowledge comes deeper and more wonderful mystery, with each answer simply leading to more questions. To Feynman, this represents the origins of the grand adventure that is science.

III. Freedom to Doubt

Perhaps the greatest value of science is the freedom it provides us to doubt. Scientific knowledge is a body of statements of varying degrees of certainty — some most unsure, some nearly sure, but none absolutely certain. This represents freedom to doubt — the idea that it is acceptable to *live* and not *know* — something that is not seen in many areas of life. Freedom to doubt is crucial to the continued development and discovery of our universe. It is the freedom to question and the freedom to think, the freedom to question the norm.

Freedom to doubt lies at the root of contrarianism and is the key that leads to the greatest progress in society.

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Suggested Reading

In Reflections on My Eighteenth Birthday – Bertrand Russell

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On the Rule of the Road – AG Gardiner

Tolerance – EM Forster

The Two Races of Men – Charles Lamb

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